Coconut Research Institute of Sri Lanka

Advisory Circular No C 4

INTERCROPPING PEPPER IN COCONUT LANDS

Pepper is one of the most profitable intercrops for coconut lands in the wet zone(Picture 1). It is a perennial intercrop and duration exceeds over 20 years. After establishment, maintenance of the crop is generally easy.

1. Selection of land

1.1.Rainfall

Most coconut lands in the wet zone (rainfall exceeding 1875 mm) in the administrative districts of Colombo, Gampaha, Kegalle, Kalutara, Galle, Matara, Matale, Kurunegala, and Puttalam are suitable. In the Wet Intermediate zone receiving an annual rainfall of 1500 - 1875 mm, pepper would require supplementary irrigation during dry months especially in the seedling and establishment stage.

Picture1: Pepper as an Intercrop

1.2.Soil

Pepper needs relative deep soils with good moisture holding capacity. Both sandy soil with low water retention capacity and hard clayey soil with poor drainage are unsuitable. A deep well-drained loamy soil with organic matter is the most suitable soil.

In slopping lands, the lower areas should be avoided because of possible water logging. In such lands the upper areas also should be avoided as they are subject to moisture stress. Hence, in slopping lands, the intermediate sections are suitable for pepper cultivation.

1.3.Light

Coconut palms should be over 15 years of age to allow sufficient sunlight for pepper. However, in underplantations, pepper could be grown on a limited scale by appropriate spacing.

Picture2: Varieties of pepper
Left *Panniyur-1*
Right - *Kuching*

2. Varieties

The introduced varieties, *Panniyur-1* and *Kuching* are recommended for coconut lands (Picture 2). However, *Panniyur-1*is the most popular among many growers. Local selections recommended by the Export Agriculture Crops Department could also be selected.

3. Planting material

Growers can obtain the planting materials from nurseries managed or approved by the Export Agriculture Crops Department. Please contact the Coconut Development Officer (CDO) for other sources of planting material. Plants with 3-4 leaves are suitable for field planting. Planting materials from unselected local vines should not be planted, as it will give rise to a low yielding plantation.

4. Planting

4.1 General guidlines

The following general guidelines should be considered:

a . To avoid competition, pepper should be planted at least 2.5 m away from coconut.

b . Pepper rows should be laid in east - west direction

Note: In attempting to intercrop coconut lands, particularly less-fertile lands, it is necessary to ensure that coconut is not adversely affected due to moisture stress and nutrient deficiency. Fertilizer should be applied to both crops regularly. The growers are requested to read the Advisory Circular: C1 for some general guidelines for intercropping in coconut lands.

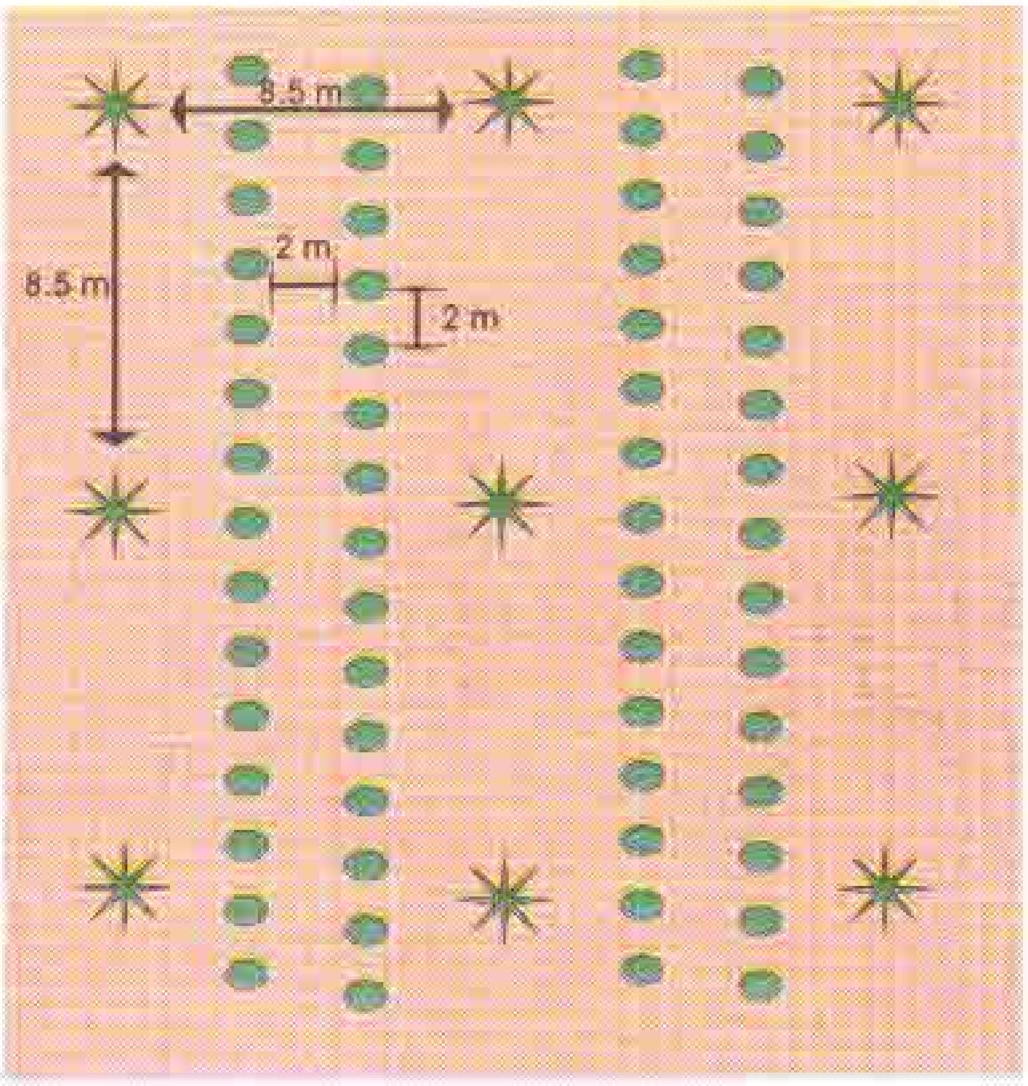
4.2 Planting system

a. Double row system

In this system two rows of pepper 2.0 m apart are planted between two rows of coconut. The pepper plants should be 2.0 m apart in the row and arranged in a triangular pattern (Picture 3) (1350 plants/ha.)

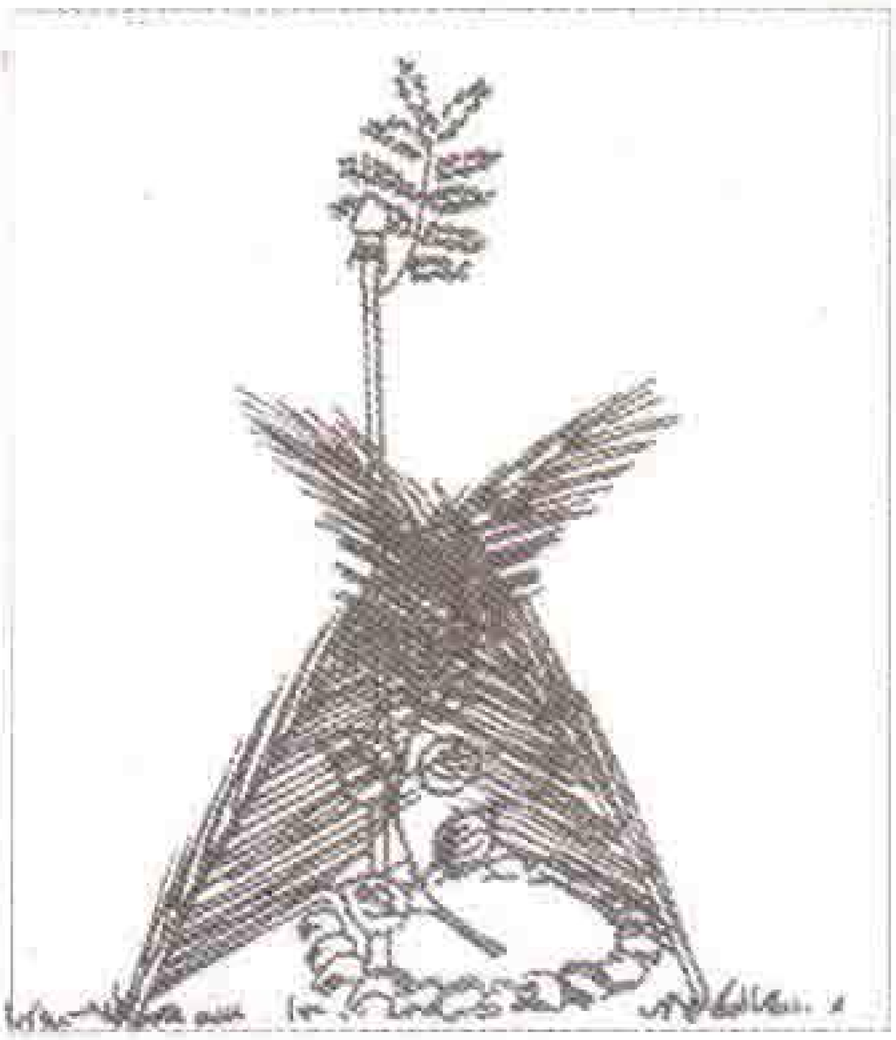
b. In order to allow movement of carts and tractors, it is advisable to plant only a single row of pepper in every five or six rows of coconut. The minimum spacing between two pepper vines should be 2.0 m.

c. In coconut plantations with irregular spacing it may not be possible to plant pepper in a systematic manner. In such lands the vines should be planted to accommodate the available space. They are at least 2.5 m away from coconut and with at least 2.0 m between two pepper vines.



Picture 3: Planting pepper in two rows

d. Pepper can also be established in 2-3 years old under plantations. A single row of pepper at a spacing of 2.0 m between vines, can be established between two rows of young palms.


Picture 4: Protect seedlings from direct sunlight

5. Field planting

Planting should be done with the onset of monsoon rains in Yala. Avoid planting in Maha as the risk of casualties is much greater. The planting holes should be 0.5m x 0.5m x 0.5m in hard soils and 0.3m x 0.3m x 0.3m in sandy loamy soils. The planting holes should be filled with top soil mixed with dry cow dung, 60g of Saphos Phosphate and few coconut husk before planting.

After planting the seedling should be protected from direct sunlight (Picture 4). The soil should be mulched with weed trash or Coconut husks.

6. Supports for pepper

Pepper needs a support for satisfactory growth and yield. Concrete posts or live supports (such as gliricidia or Kapok) could be used(Picture 5). Concrete supports tend to give a better yield of pepper, but they are expensive. Live supports are therefore popular. Mature gliricidia stakes, 3.0m long and 2.5 cm diameter should be planted about 25mm away from the pepper plant. Some wet clayey soil should be applied on cut surface at the upper end of the stake and covered with polythene to ensure better sprouting. Live supports should be planted during the monsoon proceeding the pepper planting.

Picture 5:Gliricidia as a live support for pepper

Gliricidia should be pruned to control its growth. This should be done during the rainy season. However, 1-2 small branches should be retained to assist recovery. Allow only one new shoot to develop.

7. Fertilizer application

It is important to apply fertilizer to pepper, in addition to coconut.The following mixture is recommended for pepper. The ingredients should be mixed on the farm just before use.

Urea (46%N)	4 parts by weight
Saphos Phosphate (28% P_2O_5)	5 parts by weight
Muriate of Potash (60% k_2O)	3 parts by weight
Keiserite (24% MgO)	1 parts by weight

The rate and time of application

Season	1st Year	2nd Year	3rd Year and onwords
Yala	Application 1 - 125 g Application 2 - 125 g	500 g	700 g
Maha	Application 1 - 125 g Application 2 - 125 g	500 g	700 g

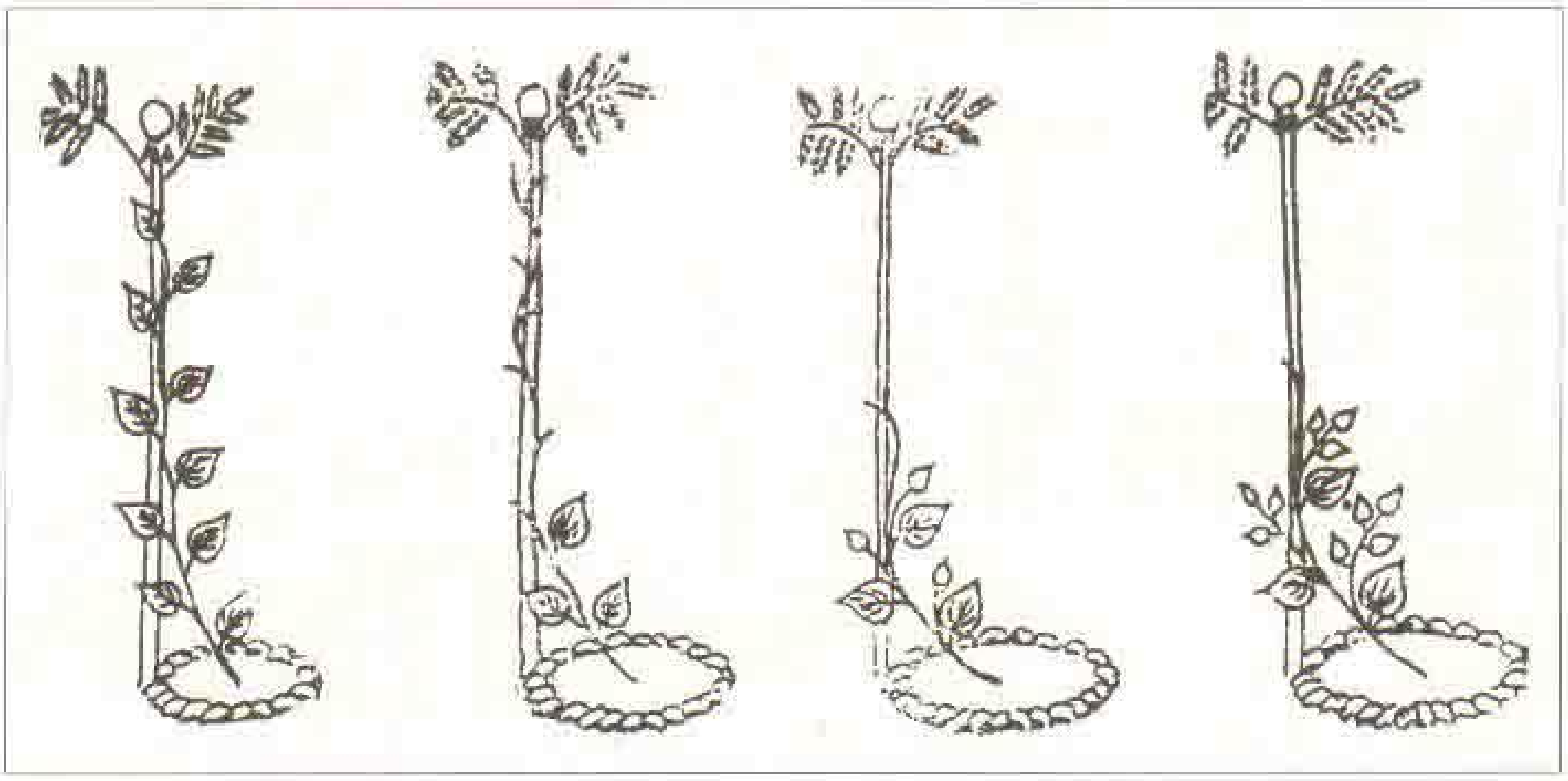
Apply fertilizer when the soil is wet and more rain is expected. Since pepper readily responds to fertilizer, split applications are recommended whenever possible. In the first year, fertilizer should be broadcast in a circular band 5 cm wide 15 cm from the plant. From second year onwards, it should be broadcast in a circular band extending from 30 cm from the plant up to 60cm. Fertilizer should be forked in and mulched well.

Yellowing of leaves due to magnesium deficiency is common in some wet areas in the southern province. This condition may be corrected by the application of 100 g of Keiserite per plant.

8. Training

When a plant has produced 8-10 leaves, remove all leaves except the lower most three leaves. After one week prune the defoliated portion of the vine. This will encourage the growth of 2 or 3 vegetative shoots. When these shoots produce 8-10 leaves each, remove all the leaves except the three basal ones of each shoot: A week later, prune the defoliate portion. This will ensure the development of more vegetative shoots. Finally each vine will have 7-9 branches and these should be tied to the support using twine or banana strands (Picture 6). The apical buds should be removed when the vines grow up to about 3 m to restrict vertical growth.

The runners that grow along the ground should be removed at the time of fertilizer application.



Picture 6: Training of a pepper vine

9. Weed control

An area up to 1 m around the plant should be clean weeded and mulched. The remaining area should be slashed. Slashing round the base of the pepper vines should be done with great care as damages to the base of the vine will either kill it or facilitate attack by pathogenic fungi.

10. Diseases

10.1. Quick wilt disease

This is caused by the fungus, *Phytophthora palmivora*. The collar rot caused by the fungus makes the leaves turn yellow and drop. Damage to the collar region and water logging during rainy season are the main predisposing factors. Affected vines should be removed and burnt. The basal portions of the health vines (about 50 cm from the ground) should be painted with Bordeaux mixture or a commonly available copper fungicides.

10.2. Little leaf disease

In this disease, abnormally small leaves are produced, which turn yellow. Yield is drastically reduced. Affected vines should be removed and burnt. Ensure that the planting materials obtain from disease-free vines

11. Pests

No major pests for pepper have so far been observed in the coconut growing areas, leaf eating caterpillar has been reported from Kegalle, Rambukkana and Mawanella areas. In such instances application of a systemic insecticide is recommended.

Thrips damage is noticeable during rainy season. Severe damage has been reported from the southern province. As a result of the insect sucking the plant sap, the leaves curl up. Then the primary buds will dry. If the damage is heavy, a systemic insecticide should be sprayed.

12. Harvesting and processing

If properly manage, pepper bears within two years of planting. About 10% of the crops is harvested during November - January. As black pepper production is popular and easy on commercial scale, fully mature but unripe (green) berries are harvested from the vines. Picking should be done carefully without damaging the tender branches. The correct stage for harvesting black pepper is when a few of the berries in the spikes are ripe. Harvested spikes are heaped up in a room for 1-2 days, and the pepper berries are separated from the spikes by trampling.Thereafter the berries are sun dried to produce black pepper. Alternately, the berries are separated and immersed for 2 minutes in hot water (80°C) and then dried either in the sun or with forced hot air. A well managed pepper vine at the age of five years will yield 2-3 kg of processed black pepper annually.

Bordeaux mixture is prepared as follows: (Use earthenware or plastic vessels)

Copper Sulphate	-	200 g
Quick lime	-	200 g
Water	-	25 litre

Dissolve Copper Sulphate overnight in 5 litres of water (suspend in a bag to facilitate dissolving). Suspend quick lime in 20 litres of water separately and stain through a fine cloth. Add Copper Sulphate to the lime, stirring vigorously. Use immediately afterwards.